

- 5 (a) State what is meant by an *ideal gas*.

A hypothetical gas whose molecules occupy negligible space and have no interactions,
and which obeys the gas laws exactly at all pressure, volume and temperature. [2]

- (b) An ideal gas, comprising single atoms, is contained in a cylinder and has a volume of $1.84 \times 10^{-2} \text{ m}^3$ at a pressure of $2.12 \times 10^7 \text{ Pa}$. The mass of the gas in the cylinder is 3.20 kg.

- (i) Determine the root-mean-square (r.m.s.) speed of the atoms of the gas.

From kinetic theory of gas,

$$KE = \frac{3}{2} nRT$$

From ideal gas equation,

$$pV = nRT$$

$$\frac{1}{2} m v_{rms}^2 = \frac{3}{2} pV \quad [1]$$

$$v_{rms} = \sqrt{\frac{3(2.12 \times 10^7)(1.84 \times 10^{-2})}{3.2}} \quad [1]$$

$$v_{rms} = 605 \text{ m s}^{-1} \quad [1]$$

root-mean-square speed = m s^{-1} [2]

- (ii) The temperature of the gas in the cylinder is 22°C .

Determine

1. the amount of gas,

Using ideal gas equation,

$$pV = nRT$$

$$n = \frac{(2.12 \times 10^7)(1.84 \times 10^{-2})}{8.31(22 + 273)} \quad [1]$$

$$n = 159 \quad [1]$$

amount = mol [2]

2. the mass of one atom of the gas.

mass of gas = number of atom $\times$ mass of atom

$$3.2 = 159 \times 6.02 \times 10^{-23} \times \text{mass of atom} \quad [1]$$

$$\text{mass of atom} = 3.34 \times 10^{-26} \text{ kg} \quad [1]$$

mass = kg [2]

- (iii) Use your answer in (ii)2. to determine the nucleon number A of an atom of the gas.

mass of atom = nucleon number $\times$ unified atomic mass

$$\text{nucleon number} = 3.34 \times 10^{-26} / 1.66 \times 10^{-27}$$

$$\text{nucleon number} = 20 \quad [1]$$

$A =$ [2]

[Total: 11]